بسم الله الرحمن الرحيم.

Section 13: triangular prism.

1). Name of theory → triangular prism منشور ثلاثي

2). Aim of experiment → 1). Achieving the minimum Limit of deviation للانحراف. العدد الأدنى

2). determine the refractive index معامل الانكسار of a triangular prism. (n)

3). used Law →

$$n = \frac{\sin\left(\frac{A+\alpha_0}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

α زاوية الانحراف ,

A زاوية رأس المنشور

4). theory of experiment → 1). Reflection Phenomenon: is the change in direction of wave front at an interface between 2 different media so that the wave front returns into the medium which it originated.

2). Refraction Phenomenon: is the change in direction of a wave passing from one medium to another or from a gradual change in medium.

3). Relative refractive index ⇒ it's ration of speed of light in one medium to speed of light in another medium.

4). Absolute Refractive index ⇒ it's ratio of light in vacuum to speed of light in another medium.

5). the angle of incidence ⇒ is the angle between a ray incident on a surface and the line perpendicular to surface at the point of incidence.

6). the angle of Refraction ⇒ is the angle between a refracted ray and the normal drawn at the point of incidence to the interface at which refraction occurs.

7). the angle of Reflection ⇒ the angle between a reflected ray and the normal drawn at the point of incidence ~~of~~ to reflecting surface.

8). Deviation angle ⇒ the acute angle between the extension of a incident ray and the extension of a exit ray.

9). The optical density: is a property of a transport material that measure speed of light through material. The extent to which any optically dense medium bends transmitted light rays toward or away from the normal.

5). tools → triangular prism, pins, white paper, Light source, protractor.

6). steps → 1). Put a triangular prism on piece of paper and define its contours.

2). Raise the prism and draw a line representing an incident ray by angle θ_1.

3). Put a number of pins on the line.

4). Look at the opposite side and we will see the image of pins formed.

5). Determine the direction of the exit ray by number of pins.

6). Raise a prism, extend the incident ray and the exit ray.

7). Determine α (deviation angle).

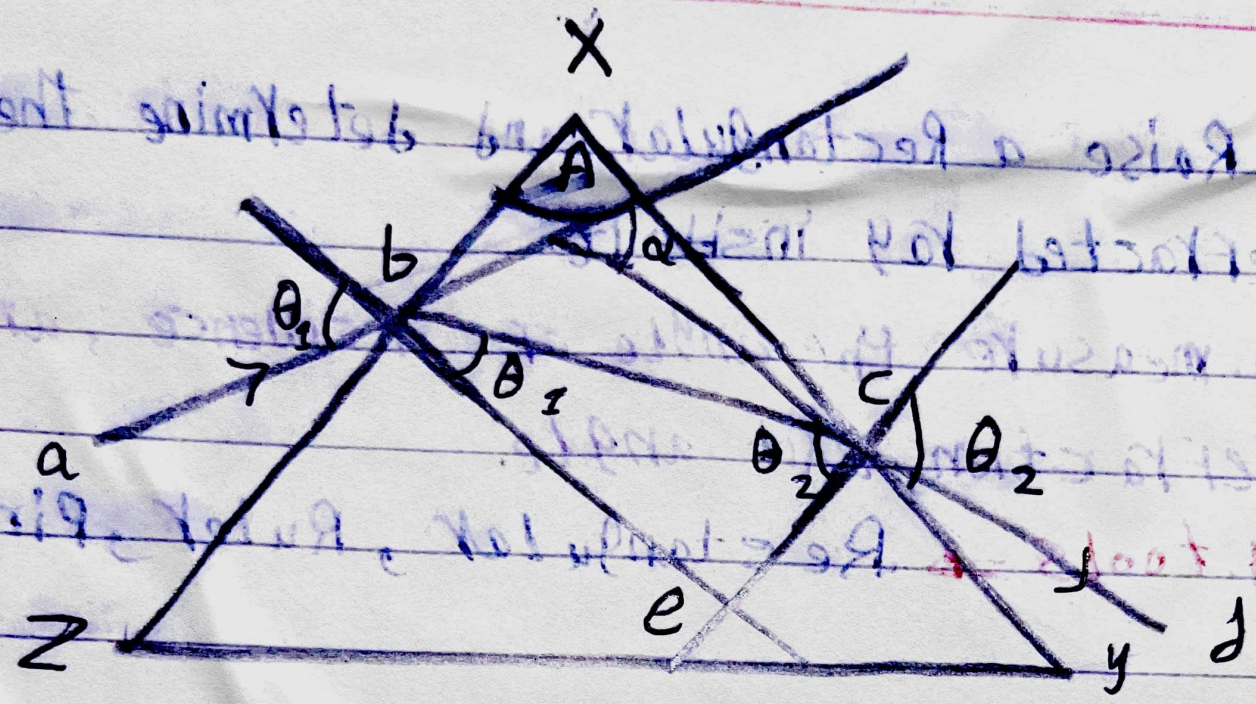
8). Repeat the previous steps until we get (α_0).

9). measure the exit angle θ_2 and calculate "n".

7). Results:-

$$n = \frac{\sin\left(\frac{A+\alpha_0}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

θ_1	α_0


X
A
b
α
θ1
θ1
c
a
θ2
θ2
e
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y
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